

Equity Portfolio Withdrawal Simulator — Methodology

How the Monte Carlo simulation works, and the data and research it rests on.

- [Equity Portfolio Withdrawal Simulator — Methodology](#)
 - [1. What is being modeled](#)
 - [2. The inputs, option by option](#)
 - [2.1 Who, and how much](#)
 - [2.2 How long, and how many](#)
 - [2.3 The return scenario](#)
 - [2.4 Spending guardrails \(upper and lower bounds\)](#)
 - [2.5 Pricing the payout](#)
 - [2.6 Inflation handling](#)
 - [3. The equity and inflation scenario model](#)
 - [3.1 Two inflation modes](#)
 - [4. Annuity pricing \(local SOA-table model\)](#)
 - [4.1 Mortality tables](#)
 - [4.2 Actuarial present value](#)
 - [4.3 Mortality improvement \(Scale G2\)](#)
 - [4.4 Calibration against the market](#)
 - [5. The withdrawal mechanism \(one path\)](#)
 - [6. Monte Carlo aggregation](#)
 - [7. Key assumptions and caveats](#)
 - [8. Dynamic inflation and interest rates](#)
 - [8.1 Specification](#)
 - [8.2 Estimation](#)
 - [8.3 Implied dynamics](#)
 - [8.4 Research basis](#)
 - [8.5 Usage and caveats](#)
 - [References](#)

1. What is being modeled

The simulator evaluates a retirement spending strategy that **keeps the money invested in equities** and, each year, withdraws **what a life annuity *would* pay** on the current balance. It is explicitly **not** a model of buying an annuity: there is no mortality pooling and no income guarantee, so a poor sequence of early returns can deplete the balance. The annuity payout is used only as a *spending rule* — a longevity-aware way to decide how much to take each year.

A single run projects one random path; the Monte Carlo driver repeats this over thousands of paths and reports the distribution of outcomes.

The pieces and their data flow:

Pricing (per-year annuity payout):

```
SOA 2012 IAM tables → annuity_pricing.py    (local, default)
immediateannuities.com → annuity_quote.py    (optional: --quotes site)
```

Scenarios (equity + inflation):

```
market_data.py          └─ US, 1928–2025
global_market_data.py ──┬─> equity_model.py    global/postwar, 1871–2020 (JST)
```

Simulation:

```
withdrawal_projection.py (one path) → montecarlo.py (many paths + CIs)
```

2. The inputs, option by option

Every input below appears in the desktop GUI, the web form, and as a command-line flag. This section explains **what each does and why it is there**; later sections give the underlying math. Defaults shown are the GUI/web defaults.

2.1 Who, and how much

- **Amount** (*default* `\$1,000,000`) — the starting invested balance, in today's dollars. Outcomes scale almost linearly with it, so it mainly sets the units.
- **Age** (*default* `65`) and **Gender** (*default* `M`) — the primary annuitant. These select the mortality table used to price the payout (Section 4): an older age implies fewer expected remaining years, so the annuity-equivalent payout *rate* is higher. Allowed ages are 40–90.
- **Joint age / Joint gender** (*default* `65 / F`) — an optional second life. When both are given the payout is priced as a **last-survivor** annuity, which pays while *either* person is alive. Because a couple's combined lifespan is longer than one person's, the payout rate is **lower** — more conservative spending — than for a single life. Leave both blank for a single-life projection; the default models a 65-year-old couple.
- **State** — used only with live **site** quotes, where insurer payouts vary by state of residence. Under the default local pricing it is ignored (use any value, or `OTHER`).

2.2 How long, and how many

- **Years** (*default* `35`) — the projection horizon. 35 years reflects planning to roughly age 100 for a 65-year-old, i.e. provisioning for **longevity** rather than just life expectancy.

- **Sims** (*default 5,000*) — the number of independent random paths. More simulations give smoother, more stable percentiles at the cost of run time.
- **Seed** (*default blank*) — the random-number seed. Blank draws fresh each run; setting an integer reproduces an identical run, which is useful for comparing one changed input at a time.

2.3 The return scenario

- **Model** (*default global*) — which historical return **sample** to draw from: **us**, **global**, or **postwar** (Section 3). In short, the US has been an outlier, so **global** and **postwar** give a more cautious, internationally-grounded picture of the future.
- **Block length** (*default 5 years*) — the length of each resampled run of consecutive years. It controls how much **sequence risk** — multi-year runs of good or bad markets, and persistent inflation — is carried into each path. A length of 1 is an IID resample with no persistence; larger blocks impose more.

2.4 Spending guardrails (upper and lower bounds)

The annuity-equivalent rule re-prices the payout on the *current* balance every year, so spending naturally rises after good markets and falls after bad ones. Two optional factors clamp that, expressed as multiples of **year 1's withdrawal in today's dollars**:

- **Upper bound** (*default 1.5*) — the maximum factor. It **caps** spending so that, even after a strong market run, you never withdraw more than 1.5× your first-year amount in real terms. Its purpose is to **preserve capital when the market does very well** rather than ratcheting consumption up with every gain — leaving more invested for later years and for a bequest. Without a cap, a few good early years pull both spending and depletion risk up sharply.
- **Lower bound** (*default blank*) — the minimum factor. It **floors** spending so that, even as the balance shrinks, withdrawals don't fall below this multiple of the first-year amount, maintaining a minimum standard of living. The trade-off is that withdrawing more than the balance can support **accelerates depletion**, so a floor raises the chance of running the account to zero.

Both clamp the **cash actually withdrawn**, so the effect feeds back into the surviving balance (Section 5). Leaving a bound blank disables it.

2.5 Pricing the payout

- **Quotes** (*default local*) — how the payout is priced: **local** uses the offline SOA mortality-table model (Section 4); **site** fetches a live quote from immediateannuities.com. Local is the default and is what the web app uses; site requires network access and a valid **State**.
- **Interest rate** (*default 4.3%*) — the discount rate used to price the payout in local mode (Section 4.2). A higher rate raises the payout, and thus the withdrawal. In constant-inflation

mode it is fixed for the whole projection; in dynamic mode it is the *starting* rate (Section 8). It has no effect under **site** quotes.

- **Scale G2 mortality improvement** (*default on; local only*) — projects the 2012 base mortality forward for expected future gains in longevity (Section 4.3). Because people are assumed to live longer, the payout rate is **lower** (more conservative).

2.6 Inflation handling

- **Inflation** (*default blank*) — in constant mode, the single assumed annual CPI rate used to express results in today's dollars (Section 3.1). Real (today's-dollar) outcomes are essentially **invariant** to this number, because each sampled year's return is restated onto the chosen basis; it mainly affects the *nominal*-dollar figures. A blank reads as 0%, and it is ignored in dynamic mode (which samples its own inflation).
- **Dynamic inflation + rate** (*default off; us sample + local only*) — instead of a constant assumption, each path uses its own sampled inflation and a discount rate that evolves with it (Section 8). It is restricted to the **us** sample because the rate model is calibrated on US data; it is disabled for the global/postwar samples.

3. The equity and inflation scenario model

Each projected year needs a **nominal equity return** (to grow the account) and an **inflation rate** (to express results in today's dollars). Because equity returns and inflation are correlated, they are drawn **jointly** rather than independently — a sampled year always carries both numbers together.

Data sources. Two paired annual datasets are available:

- **United States, 1928–2025** (98 years): S&P 500 nominal total return (dividends reinvested), from Damodaran / NYU Stern [4], paired with CPI-U annual-average inflation [5].
(`market_data.py`)
- **Broad developed markets, 1871–2020** (16 countries that have equity data; ~2,260 country-years): per-country nominal equity total return and CPI inflation from the **Jordà–Schularick–Taylor Macrohistory Database** [14] (`global_market_data.py`). The US has been an *ex post* outlier among developed markets — its realised returns sit near the top of the cross-section and its worst drawdowns are mild next to the war- and hyperinflation-era losses other countries suffered — so a forward-looking distribution is arguably better drawn from many markets than from the US alone [15].

Sampling. All samples use a **circular block bootstrap** [3]: consecutive runs of `block_length` years (default 5) are resampled and stitched together, wrapping around the series end. Versus an IID resample this preserves **serial correlation** — momentum, mean reversion, and multi-year runs of high or low inflation — which is what makes *sequence-of-returns* risk show up and which widens the left tail of multi-year outcomes. (Historical resampling is preferred over normal-based Monte Carlo for retirement risk work, which understates tail risk and overstates safe withdrawal rates [1,

2].) For the multi-country samples each block is drawn from a **single country** (countries weighted by their number of years), so within-country sequence risk is preserved while the cross-section of national outcomes enters the distribution.

Three return **samples** are offered (`equity_model.py` , the "Model" input):

- **us** — the US series alone (1928–2025).
- **global** — the full broad developed-market sample (1871–2020). The most cautious: it includes catastrophic equity outcomes the US never experienced (e.g. Germany and Japan around WWII, with real one-year losses near –90%).
- **postwar** — the same broad sample restricted to **1950 and later**. The rationale is that the post-WWII global order (Bretton Woods and its successors, no great-power war among developed economies, modern independent central banking) is a more relevant guide to the future than the 1870–1945 era of world wars and hyperinflations. It sits between **us** and **global** — broader than the US, but without the pre-war disasters.

Each report prints summary statistics for the chosen sample (in **real** terms, so hyperinflation years do not distort them).

3.1 Two inflation modes

The simulator supports two ways of handling inflation:

- **Constant** (default). Inflation is a single assumption (`--inflation` , default **2.5%**), used only to deflate results to today's dollars, and the annuity discount rate is fixed (`--interest` , default 4.3%). To keep the equity series consistent with the constant assumption, each sampled year's *nominal* equity return is **restated** onto that basis — its embedded historical inflation is stripped out (preserving the real return) and the constant rate re-applied:

```
real = (1 + nominal) / (1 + historical_inflation) - 1
restated = (1 + real) × (1 + assumed_inflation) - 1
```

so the historical inflation once correlated with equity has no residual effect.

- **Dynamic** (`--dynamic-rates`). Inflation is *not* held constant: each path uses its own per-year sampled inflation directly (from the block bootstrap, so persistence is preserved), it deflates results to today's dollars, **and** it drives an evolving annuity discount rate. Equity is used as drawn (no restatement), so the historical equity/inflation pairing stays intact. Section 8 specifies and estimates this mode. Dynamic mode is available **only with the us sample** (and local pricing): the discount-rate model is calibrated on US data, so evolving it from a foreign — possibly hyperinflation — inflation path is not meaningful. The `global` and `postwar` samples therefore always use the constant-inflation mode (their results are real, and real outcomes are invariant to the constant chosen).

4. Annuity pricing (local SOA-table model)

By default the per-year payout is priced **offline** from published mortality tables (`annuity_pricing.py`); live quotes from `immediateannuities.com` [6] are an opt-in alternative (`--quotes site`).

4.1 Mortality tables

The base mortality is the **Society of Actuaries 2012 Individual Annuity Mortality (IAM) Basic Table**, by sex, age-nearest-birthday [7]:

- `soa_mortality_2581.csv` — *2012 IAM Basic Table – Male, ANB (ages 0–120)*
- `soa_mortality_2582.csv` — *2012 IAM Basic Table – Female, ANB (ages 0–120)*

These are **annuitant** tables: annuitants are healthier and longer-lived than the general population. Optional generational mortality improvement uses **SOA Projection Scale G2** [7]:

- `soa_mortality_2583.csv` — *Projection Scale G2 – Male, ANB (ages 0–105)*
- `soa_mortality_2584.csv` — *Projection Scale G2 – Female, ANB (ages 0–105)*

4.2 Actuarial present value

With a flat interest rate `i` and survival probabilities `tpx` (the probability a life now aged `x` survives `t` years, built from the table's one-year mortality rates `q_x`), the actuarial present value of `\$1` per year of income paid **monthly, in arrears, while alive** is [8]:

$$\begin{aligned} v &= 1 / (1 + i) \\ a_x &= \text{sum over } t \geq 1 \text{ of } v^t \cdot tpx && \text{(annual annuity-immediate)} \\ a_x^{(12)} &\approx a_x + 11/24 && \text{(Woolhouse 2-term, } m = 12) \end{aligned}$$

The premium buys income at that price, so the **level annual payout** is

$$\text{annual_payout} = \text{premium} / a_x^{(12)}.$$

For two lives, a **last-survivor** annuity pays while *either* is alive; its factor follows from inclusion–exclusion [8],

$$a_{LS} = a_x + a_y - a_{xy},$$

where `a_xy` (both alive) uses the product of the two independent survival curves, with the same `+11/24` monthly adjustment. The table's top age (120) is treated as certain death, so the survival

curve terminates cleanly.

4.3 Mortality improvement (Scale G2)

With `--improvement`, the 2012 base rates are projected forward **generationally** to the quote year [7]:

$$q_a(\text{year}) = q_a(2012) \cdot (1 - g2_a)^{(\text{year} - 2012)},$$

applied as each cohort ages, so the calendar year at attained age `a` is `quote_year + (a - start_age)`. Improvement lengthens lifespans and therefore *lowers* the payout. (Scale G2 grades to zero by its terminal age; attained ages beyond it receive no further improvement.)

4.4 Calibration against the market

The model carries **no insurer expense, profit, or interest-rate load**, so it is a clean actuarial benchmark rather than a marketed quote. Compared against live immediateannuities.com quotes [6] for ages 60/65/70/80, the model pays roughly **80–94%** of the site at 3.5% interest and **matches the site at about 5%** interest; at the **default 4.3%** it sits between these, slightly below parity. That ~5% is consistent with a current 10-year nominal yield of roughly a 2.3% real rate plus ~2.5% inflation (Section 8).

5. The withdrawal mechanism (one path)

Starting from the invested `amount`, for each projected year (`withdrawal_projection.py`):

1. Price the monthly annuity payout per dollar at the current age and balance (Section 4); the payout is linear in premium, so one rate per age is computed and scaled.
2. Withdraw **12 ×** that monthly payout from the balance.
3. Grow the remaining balance by that year's nominal equity return.
4. Increment the age(s) by one and repeat.

Optional `--upper-bound` / `--lower-bound` cap and floor the annual withdrawal **in today's dollars**, as factors of year 1's withdrawal (e.g. cap at 1.2× and floor at 0.5×); the clamp is applied to the cash actually withdrawn, so it feeds back into the surviving balance.

6. Monte Carlo aggregation

`montecarlo.py` runs many independent paths (default **5,000**) and summarizes the distribution:

- **Ending balance**, in both nominal and today's dollars: mean, median, and the central **80% / 90% / 95% / 99%** confidence intervals. A "C% interval" is the central range covering C% of

outcomes — the $[(100-C)/2, 100-(100-C)/2]$ percentile band (e.g. 80% = 10th–90th percentile).

- **Annual payout by year**, in today's dollars: the same statistics, per year.
- **Downside**: the share of paths whose real ending balance finishes below the starting amount, and below half of it.
- **Worst real returns**: the worst single-year and worst cumulative five-year *real* (inflation-adjusted) equity total return seen anywhere in the run. This is the **market** return itself, before any withdrawal — not the change in the account balance, which falls by more in a down year because spending is also taken out.

The annuity-rate cache is built once per run and reused across all paths, so pricing cost (and any network traffic) stays at roughly one rate per age regardless of the number of simulations.

7. Key assumptions and caveats

- This models withdrawing the annuity-*equivalent* amount while staying invested. It is **not** an annuity purchase — no mortality pooling, no income guarantee. Because the full annuity payout includes return of principal, the withdrawal rate is high and the real balance typically erodes; single-life runs commonly finish below the starting amount in real terms. This is expected model behavior, not a defect.
- Local pricing omits insurer loads, so it is an actuarial benchmark, not a quotable rate; results are sensitive to the assumed interest rate.
- Historical calibration assumes the 1928–2025 distribution is informative about the future, which may not hold.
- Mortality follows population-level annuitant tables; no individual health underwriting is performed.

8. Dynamic inflation and interest rates

The dynamic mode (`--dynamic-rates` , `rate_model.py`) makes inflation vary year to year and ties the annuity discount rate to it, so a higher-inflation path produces higher nominal yields and a larger nominal annuity payout. Because inflation and bond yields are persistent, correlated time series, the discount rate is modelled with **lagged** dependence rather than as a static function of the current year's inflation.

8.1 Specification

The design is **layered**: inflation keeps coming from the block bootstrap of Section 3 (which already preserves its serial persistence and its pairing with equity), and only the interest rate is added as a parametric process. The rate partially adjusts each year toward a long-run **Fisher target** driven by the *previous* year's inflation:

$$i^*_t = a + b \cdot \pi_{t-1} \quad (\text{long-run Fisher target})$$

$$i_t = (1 - \lambda) \cdot i_{t-1} + \lambda \cdot i^*_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

a is the real-rate intercept, b the long-run inflation pass-through, and λ the annual speed of adjustment. The rate in year t therefore depends on **both** the prior year's rate (weight $1 - \lambda$) and the prior year's inflation. Year 1 is priced at a given initial ("today's") rate i_0 (`--initial-rate`, default 4.3% $\approx$ the current 10-year yield); the rate is floored at zero. The interest innovation is drawn independently of the equity/inflation draws — a deliberate simplification (see §8.5).

8.2 Estimation

The model is fit as the reduced form

$$i_t = c + \rho \cdot i_{t-1} + \delta \cdot \pi_{t-1} + \varepsilon_t$$

by ordinary least squares, then mapped to the structural parameters via $\lambda = 1 - \rho$, $a = c / \lambda$, $b = \delta / \lambda$. The data are annual averages of the **10-year Treasury constant-maturity yield** (Federal Reserve / FRED series `GS10`, which begins in 1953) [11] and **CPI-U inflation** [5]; the usable sample is **1954–2025 (n = 72)**. `rate_model.fit_interest_model()` reproduces this fit (the script `fit_rate_model.py` prints it and regenerates the focused fit report `FIT.md` / `FIT.pdf`), and the package defaults are exactly these estimates.

Exhibit 1 — OLS estimates (dependent variable: 10-year yield i_t):

Term	Coefficient	Std. error	t-stat
constant c	0.00279	0.00202	1.38
i_{t-1} (ρ)	0.83477	0.04238	19.70
π_{t-1} (δ)	0.18645	0.04478	4.16

$n = 72$ (1954–2025) · $R^2 = 0.927$ · adj. $R^2 = 0.925$ · residual $\sigma = 0.79$ pp · Durbin–Watson = 1.95.

Mapped to structural parameters:

$$\lambda = 1 - \rho = 0.165 \quad (\text{annual adjustment speed})$$

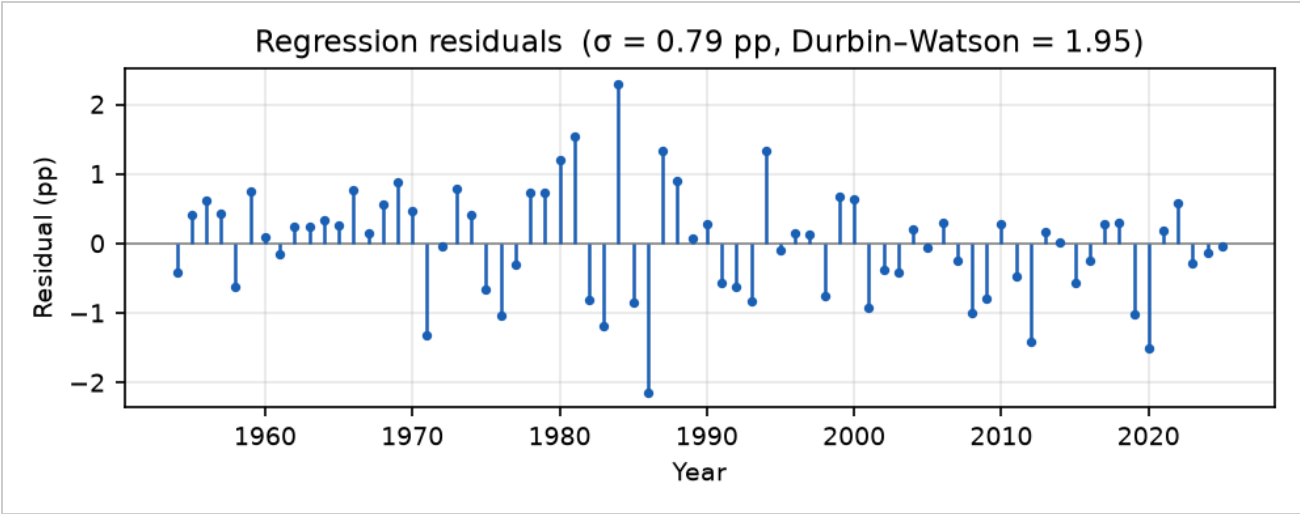
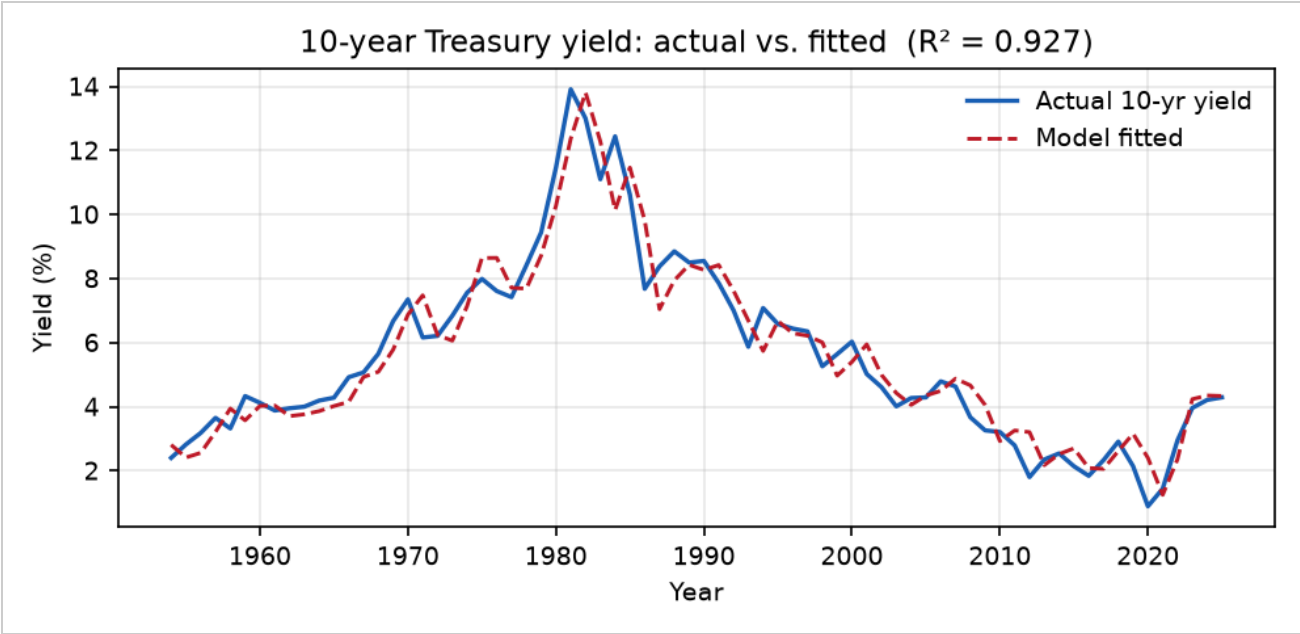
$$a = c / \lambda = 1.69\% \quad (\text{real-rate intercept})$$

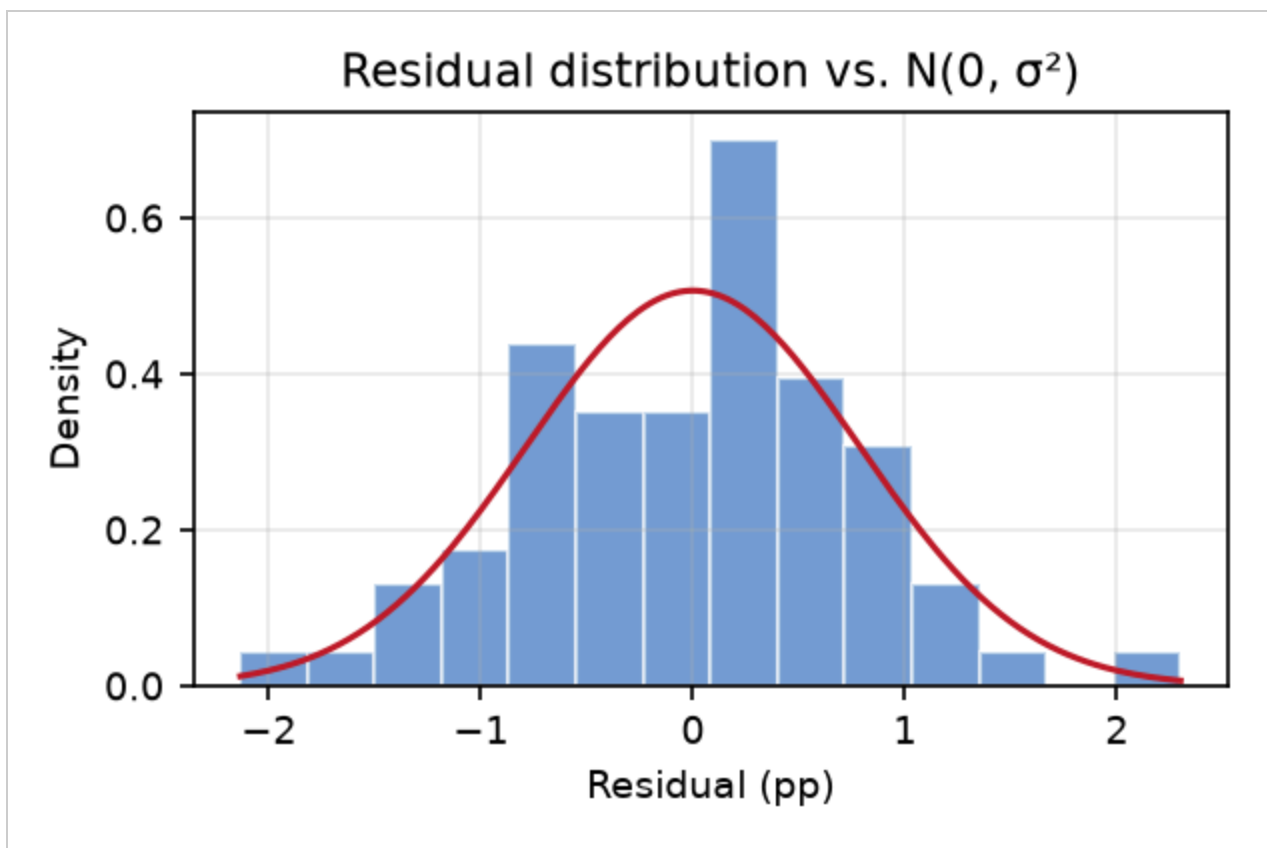
$$b = \delta / \lambda = 1.13 \quad (\text{long-run inflation pass-through})$$

$$\sigma = 0.79 \text{ pp} \quad (\text{innovation standard deviation})$$

The lagged yield is overwhelmingly significant ($t \approx 20$) and the lagged-inflation term is significant ($t \approx 4.2$); the model explains 93% of the variation in the yield, and the Durbin–Watson near 2 indicates

essentially no residual autocorrelation — the lagged specification has absorbed the dynamics.



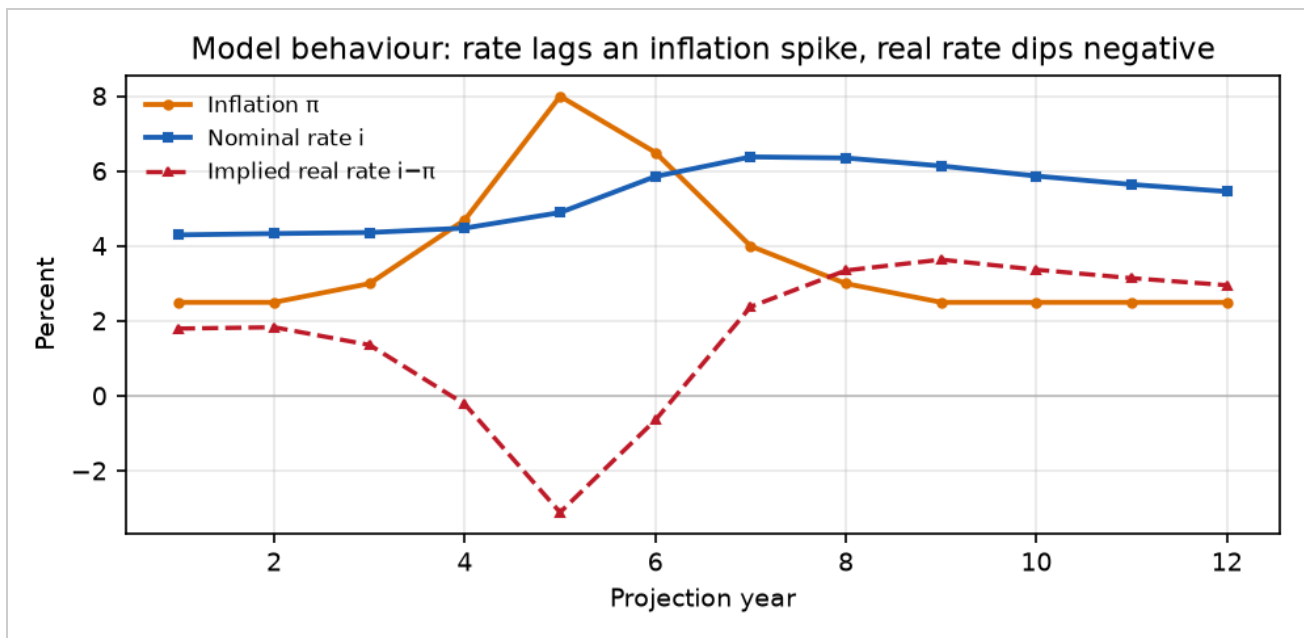


8.3 Implied dynamics

The estimates have a clear economic reading:

- **Long-run pass-through** $b \approx 1.13$ is slightly above one, consistent with the tax-augmented (Darby) form of the Fisher hypothesis.
- **Adjustment speed** $\lambda \approx 0.165$ is *small*: only about one-sixth of the gap to the Fisher target closes each year. This is the sluggish short-run pass-through (the Mundell–Tobin / "Fisher puzzle" effect, §8.4).
- The **long-run real rate is** $\approx a = 1.7\%$, and the steady-state nominal rate at 2.5% inflation is $a + b \cdot 2.5\% \approx 4.5\%$ — close to both the current 10-year yield and the ~5% that reproduces live annuity quotes (§4.4).

Crucially, because λ is small, when inflation jumps the rate lags and the **implied real rate** $i_t - \pi_t$ **compresses and can go temporarily negative**, recovering only as the rate catches up — exactly the 2021–2022 experience — while the long-run real rate stays near 1.7%. Exhibit 4 illustrates this with a deterministic inflation spike.



8.4 Research basis

The specification follows established findings on the yield/inflation link:

- **Fisher relationship [9].** A nominal yield $\approx$ expected real rate + expected inflation + term premium; over long horizons the pass-through is roughly one-for-one (cointegration evidence across OECD countries) — consistent with the fitted $b \approx 1.1$.
- **Sluggish short-run pass-through (Mundell–Tobin / "Fisher puzzle") [10].** In the short run the pass-through is well below one, so when inflation jumps, nominal yields lag and the real rate compresses — captured here by the small fitted λ and the lagged inflation term.
- **The real rate is not constant — and not always positive [11, 12, 13].** The 10-year TIPS (market real) yield was negative from December 2011 through ~2013 and deeply negative in 2020–2022 (a record-low -0.93% 10-year TIPS auction in July 2020, around -1.2% in 2021). On a realized basis the divergence was extreme: with CPI near 7% at end-2021 and a 9.1% peak in mid-2022 against a 10-year yield of $\sim 1.5\text{--}3\%$, the **ex-post real yield was roughly -5% to -6%** . It is about $+2.3\%$ as of mid-2026. The error-correction model reproduces this behavior endogenously while remaining mean-reverting, so it cannot drift to implausible levels over a long horizon.

8.5 Usage and caveats

Enable with `--dynamic-rates` (local pricing only); set the starting rate with `--initial-rate`.
Caveats specific to this mode:

- The interest innovation ε_t is drawn **independently** of the equity and inflation draws. Contemporaneous correlation between rate shocks and equity/inflation is therefore omitted; this is a simplification, not an estimate of zero correlation.

- Parameters are fit on **1953–2025** (the span of the GS10 series), a shorter window than the 1928–2025 equity/inflation data, and assume that historical yield/inflation relationship continues to hold.
- The model is of the **nominal** rate; the "larger payout" from higher inflation is largely a nominal effect, and is deflated away when results are expressed in today's dollars (a level annuity loses real value faster under high inflation).

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